

ANTIPAYUTA, Russian Federation

WMO: 230580

Lat: 69.104N

Lon: 76.858E

Elev: 11

StdP: 14.69

Time Zone: 5.00 (E05)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -46.2	(c) -41.9	(d) -50.7	(e) 0.3	(f) -45.0	(g) -47.1	(h) 0.4	(i) -41.2	(j) 34.7	(k) 14.2	(l) 30.8	(m) 12.7	(n) 5.8	(o) 60	(p) 0.954

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.9	(c) 76.8	(d) 64.9	(e) 70.7	(f) 61.6	(g) 65.2	(h) 58.5	(i) 65.8	(j) 74.1	(k) 62.7	(l) 70.8	(m) 58.9	(n) 64.6	(o) 6.9	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 62.4	(b) 84.4	(c) 71.0	(d) 58.9	(e) 74.3	(f) 67.0	(g) 55.8	(h) 66.3	(i) 62.7	(j) 30.6	(k) 73.9	(l) 28.3	(m) 71.6	(n) 25.8	(o) 64.4	(p) 74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 28.7	(b) 25.1	(c) 22.2	(e) -47.5	(f) 81.5	(g) 6.9	(h) 5.1	(i) -52.5	(j) 85.2	(k) -56.5	(l) 88.2	(m) -60.4	(n) 91.0	(o) -65.4	(p) 94.7		
(4) 28.7	(5) 25.1	(6) 22.2	(7) -46.7	(8) 67.3	(9) 5.1	(10) 4.4	(11) -50.3	(12) 70.5	(13) -53.3	(14) 73.1	(15) -56.1	(16) 75.5	(17) -59.8	(18) 78.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.9	-15.1	-13.0	-1.4	7.8	22.9	43.2	55.8	48.0	39.6	21.5	-0.6	-8.0		
	DBStd	27.39	15.65	17.73	16.53	14.42	8.89	9.38	9.22	5.68	6.48	12.67	14.74	16.45		
	HDD50	12398	2018	1764	1594	1267	840	250	40	109	317	883	1519	1797		
	HDD65	17591	2483	2184	2059	1717	1305	654	319	528	763	1348	1969	2262		
	CDD50	317	0	0	0	0	0	46	219	47	5	0	0	0		
	CDD65	33	0	0	0	0	0	0	33	0	0	0	0	0		
	CDH74	268	0	0	0	0	0	1	267	0	0	0	0	0		
	CDH80	55	0	0	0	0	0	0	55	0	0	0	0	0		
	WSAvg	10.7	10.8	10.9	11.2	11.5	12.6	10.5	10.1	10.1	9.7	10.0	10.2	10.7		
	PrecAvg	9.8	0.5	0.5	0.6	0.6	0.6	1.0	1.0	1.5	1.2	1.0	0.7	0.6		
	PrecMax	20.0	1.3	1.3	1.7	1.7	1.2	5.5	5.2	3.5	2.5	2.0	1.9	1.8		
(15) Precipitation	PrecMin	4.1	0.1	0.0	0.1	0.0	0.1	0.2	0.0	0.1	0.1	0.3	0.1	0.2		
	PrecStd	3.7	0.3	0.4	0.4	0.4	0.4	1.0	1.1	1.1	0.6	0.4	0.4	0.4		
	DB	25.0	26.0	32.2	35.9	44.3	72.1	84.6	66.9	59.7	46.9	32.3	27.7			
	MCWB	24.5	25.2	31.6	33.7	41.5	60.4	68.6	59.2	54.0	45.3	32.0	27.2			
	2% DB	20.9	21.8	28.9	34.0	38.4	67.3	81.2	63.3	55.2	41.4	28.9	24.2			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	20.5	21.3	28.3	32.7	36.3	58.3	66.7	58.0	50.5	39.9	28.6	23.7			
	DB	15.4	18.1	25.7	32.6	35.8	62.6	76.4	60.1	51.7	38.8	25.7	20.9			
	MCWB	14.9	17.5	25.0	31.7	34.3	54.8	64.8	55.8	48.6	37.5	25.3	20.3			
	DB	6.8	13.6	22.5	30.4	34.2	57.6	71.4	57.5	49.2	35.7	21.9	16.7			
	MCWB	6.2	13.0	21.9	29.7	33.1	51.8	62.5	54.0	46.9	34.7	21.4	16.1			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	24.5	25.5	31.6	34.1	41.4	62.4	71.6	61.5	55.2	45.1	32.1	27.1			
	MCDB	24.7	25.9	31.9	35.0	44.0	70.1	82.1	65.3	58.5	46.3	32.2	27.5			
	WB	20.5	21.3	28.3	33.0	36.4	58.8	68.5	58.8	51.3	40.3	28.7	23.7			
	MCDB	21.0	21.7	28.7	33.9	38.0	65.9	77.8	62.3	54.5	41.2	28.9	24.2			
	WB	15.2	17.6	25.1	31.9	34.5	55.5	65.6	56.5	49.1	37.5	25.4	20.4			
(31) Mean Daily Temperature Range	MCDB	15.6	18.1	25.5	32.6	35.4	61.7	73.8	59.4	51.4	38.7	25.7	20.9			
	WB	6.5	13.1	21.9	29.6	33.2	52.6	63.3	54.5	47.1	34.8	21.4	16.2			
	MCDB	6.9	13.8	22.5	30.4	34.1	57.2	71.4	57.0	48.9	35.6	22.0	16.7			
	MDBR	12.6	13.9	15.2	16.6	10.1	11.0	13.9	10.6	8.7	9.4	12.7	14.6			
	5% DB	18.7	16.1	16.3	12.2	8.1	19.0	19.2	14.2	12.9	6.6	13.3	15.8			
(35) Clear-Sky Solar Irradiance	MCWBR	18.8	16.1	16.3	12.1	6.9	11.6	8.3	9.6	9.2	5.9	13.2	15.5			
	MCDBR	18.6	16.1	16.2	12.8	7.4	18.0	18.5	12.8	11.3	6.2	12.9	15.9			
	MCWBR	18.6	16.1	16.3	12.8	6.5	12.1	9.3	9.8	8.8	5.7	13.0	15.6			
	taub	0.209	0.248	0.301	0.321	0.339	0.359	0.386	0.369	0.317	0.252	0.198	N/A			
	taud	1.867	1.981	2.024	2.066	2.124	2.345	2.329	2.356	2.429	2.151	1.901	N/A			
(42) All-Sky Solar Radiation	Ebn at Noon	9	153	218	255	266	265	251	241	226	155	8	0			
	Edh at Noon	2	15	28	37	39	33	32	28	20	12	1	0			
(44) All-Sky Solar Radiation	RadAvg	5	101	572	1254	1721	1676	1584	1014	512	157	20	0			
	RadStd	0	17	62	110	102	138	202	138	59	23	2	0			

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		+2.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page